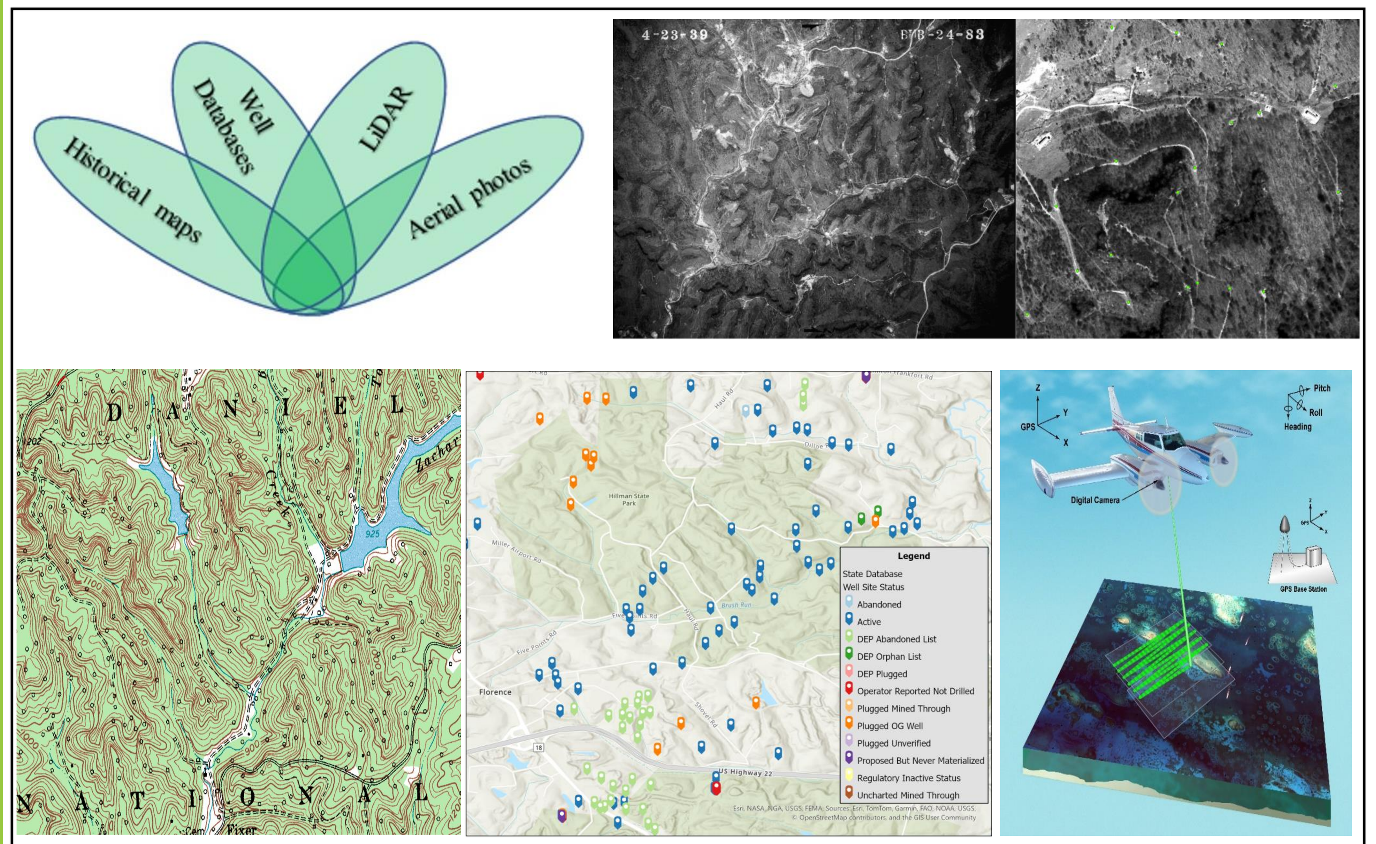
Extended Application of State LiDAR Datasets in Locating Orphaned Wells in the Appalachian Region

V. Raghava Gorantla[1,2]; James Sams[1,2]; Matthew Reeder[1]; Natalie Pekney[1]
[1]National Energy Technology Laboratory, 626 Cochran Mill Road, Pittsburgh, PA 15236, USA; [2]NETL Support Contractor, 626 Cochran Mill Road, Pittsburgh, PA 15236, USA

Abstract

Orphaned oil and gas wells pose a significant environmental threat to surrounding ecosystems and groundwater quality. Location inaccuracies in historical and state oil and gas well databases present a major challenge in locating these orphaned wells. To address this, modern scientific methods such as Light Detection and Ranging (LiDAR), aerial magnetic remote sensing, and digital GIS products have been employed. LiDAR technology uses light to detect surface area changes, providing detailed surface views. Topographic LiDAR can detect changes in surface elevation with a vertical accuracy of 10 cm, and the Digital Elevation Models (DEMs) generated from LiDAR Point Cloud (LPC) data offer high-resolution terrain representations. LiDAR surveys, which can be conducted via aircraft or small Unmanned Aerial Systems (sUAS), are more commonly aircraft-based when state-funded due to cost and time efficiency, with data often publicly accessible. Although these surveys are not initially intended for locating orphaned wells, the trails identified from DEMs can be used as access roads leading to well sites in combination with other well data resources. The National Energy Technology Laboratory (NETL) has developed a LiDAR processing workflow to fully utilize state LiDAR data to identify trails and access roads to well sites that are not visible through aerial or satellite imagery. This tool significantly improves the efficiency of locating orphaned wells. NETL's efforts in processing DEMs from state LiDAR data have been applied in orphaned well-finding initiatives in the Appalachian region covering Pennsylvania, New York, Kentucky, West Virginia, and Ohio.



Data Resources.

Well Site Infrastructure.

Advantages	Challenges
No cost associated with obtaining the data since the data has been already collected by the state GIS agencies.	Age of the LiDAR point cloud data is crucial, e.g., data collected after 2014 yields best results.
Works well in locating orphaned oil and gas wells especially densely forest areas in the Appalachian region.	Lapse in spatial resolution since the data has been collected via aircraft to identify old wells (>50 years) especially if there was a surface disturbance.

Acknowledgement
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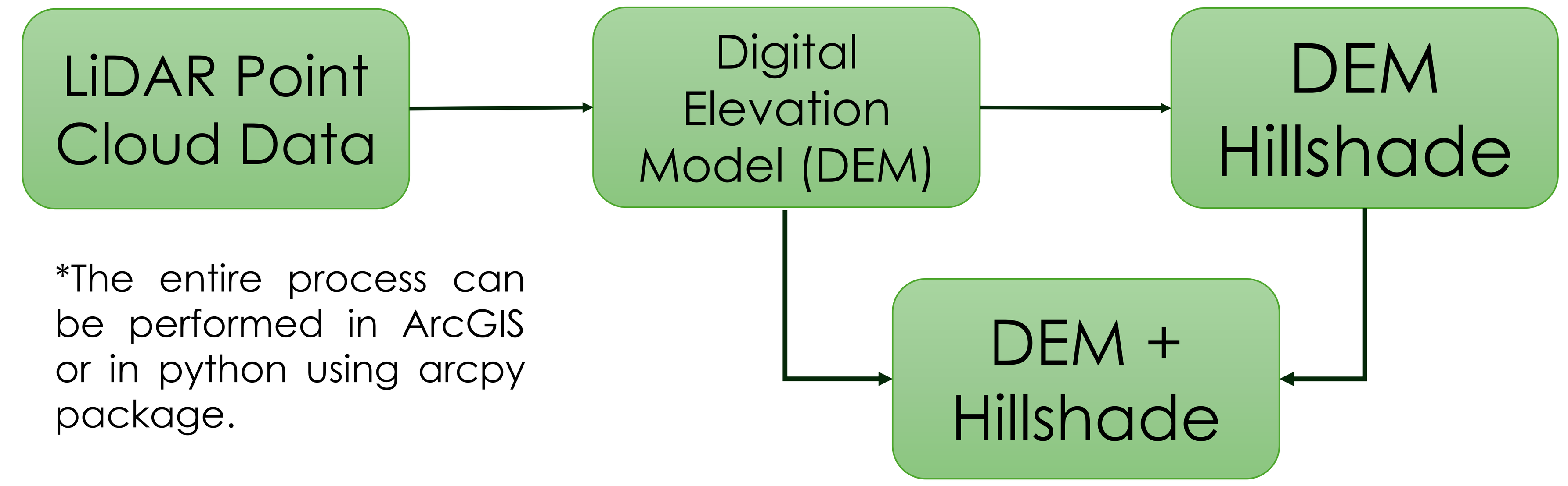
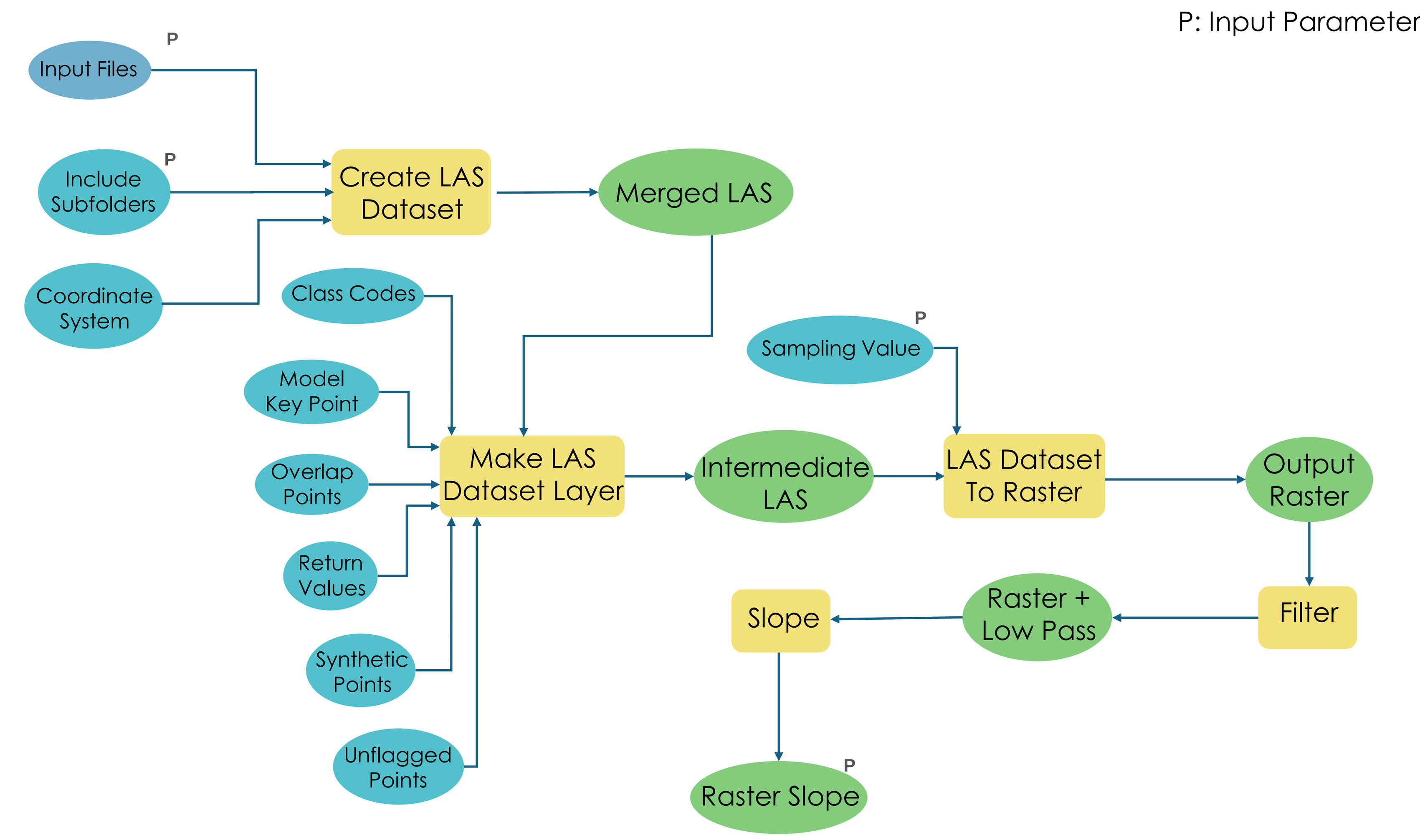
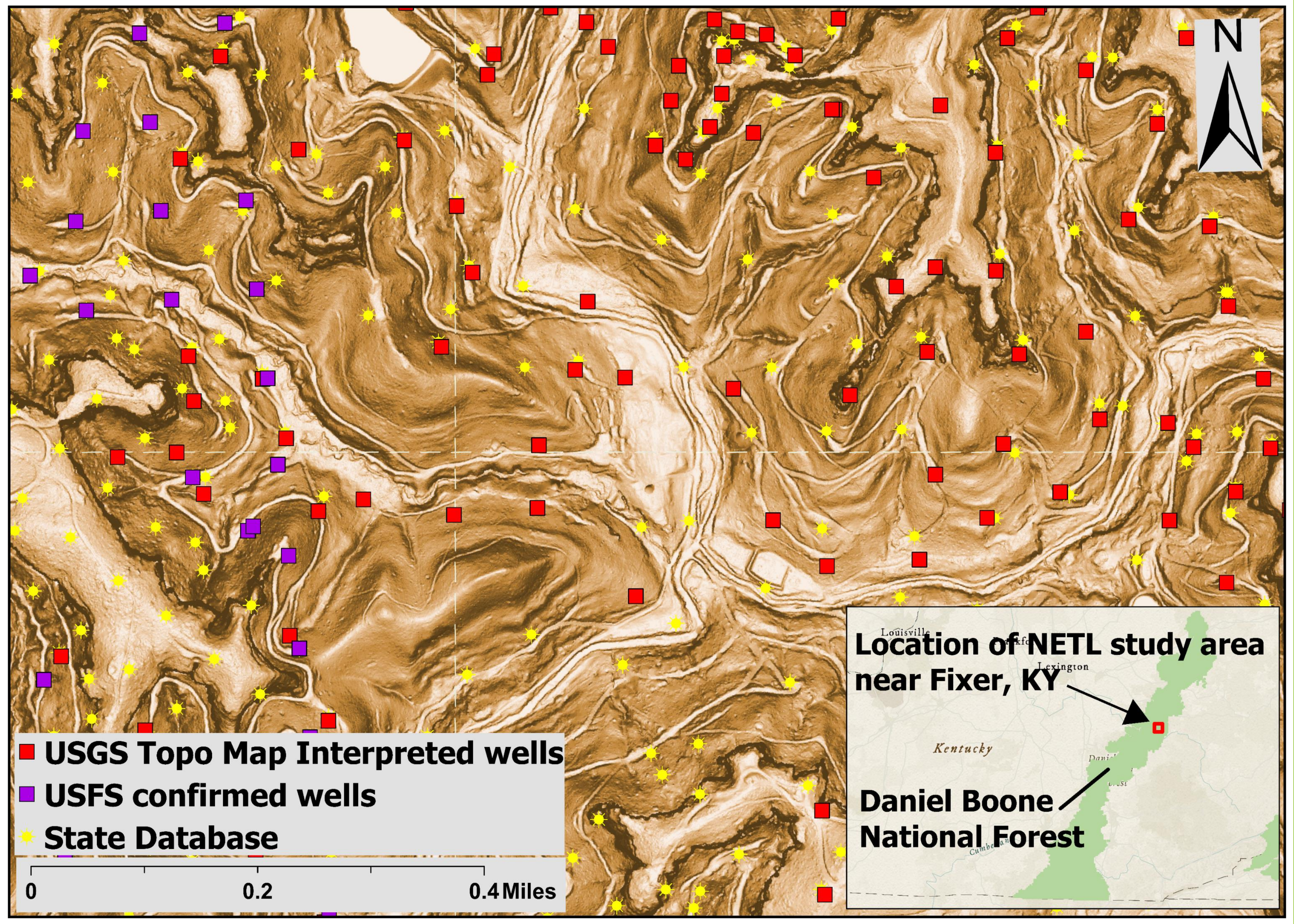
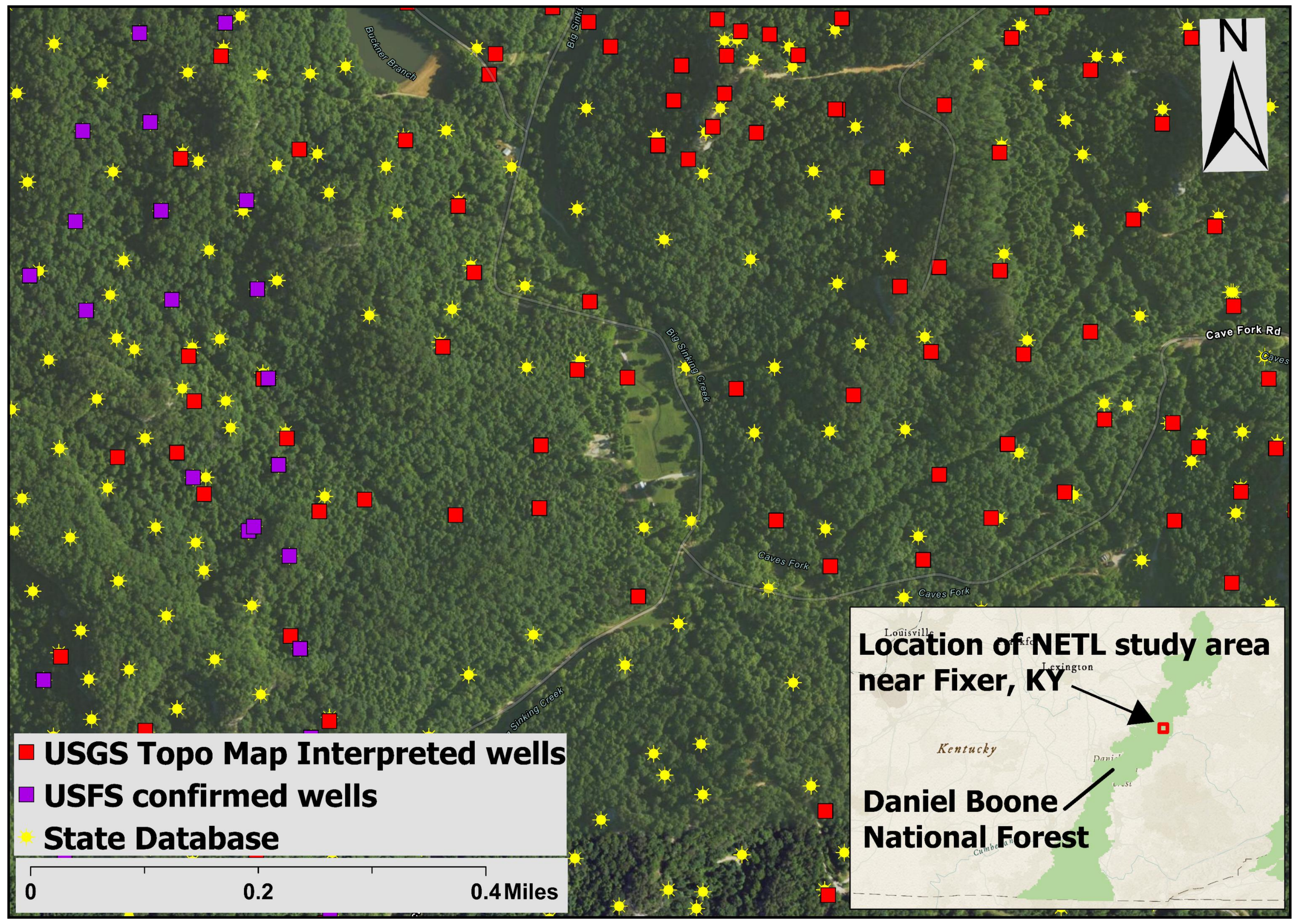


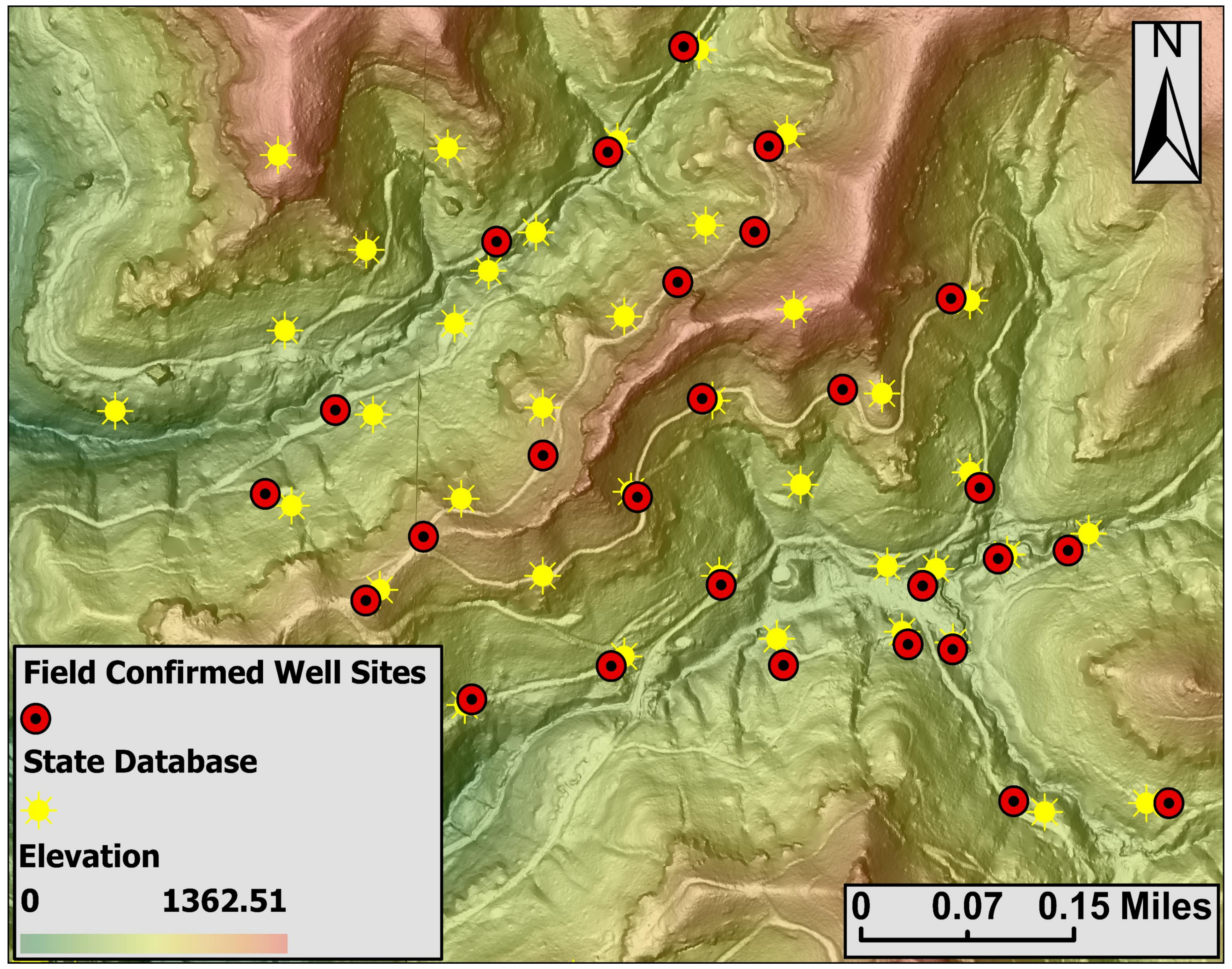
LiDAR Data to DEM Workflow.



LiDAR Point Cloud Data Processing Workflow.







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